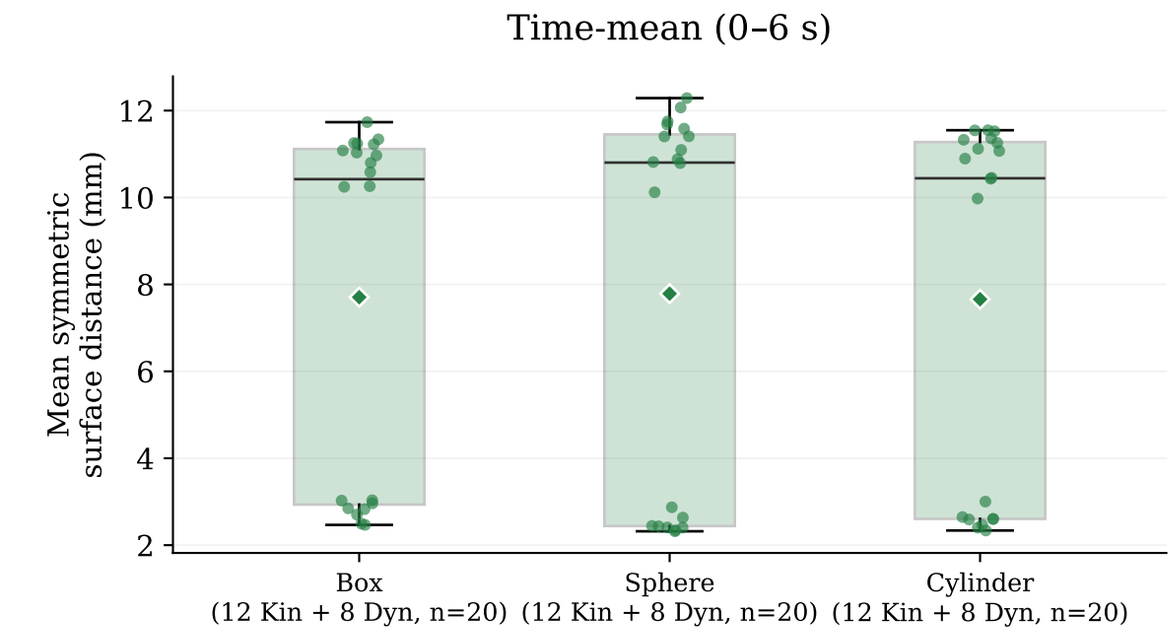
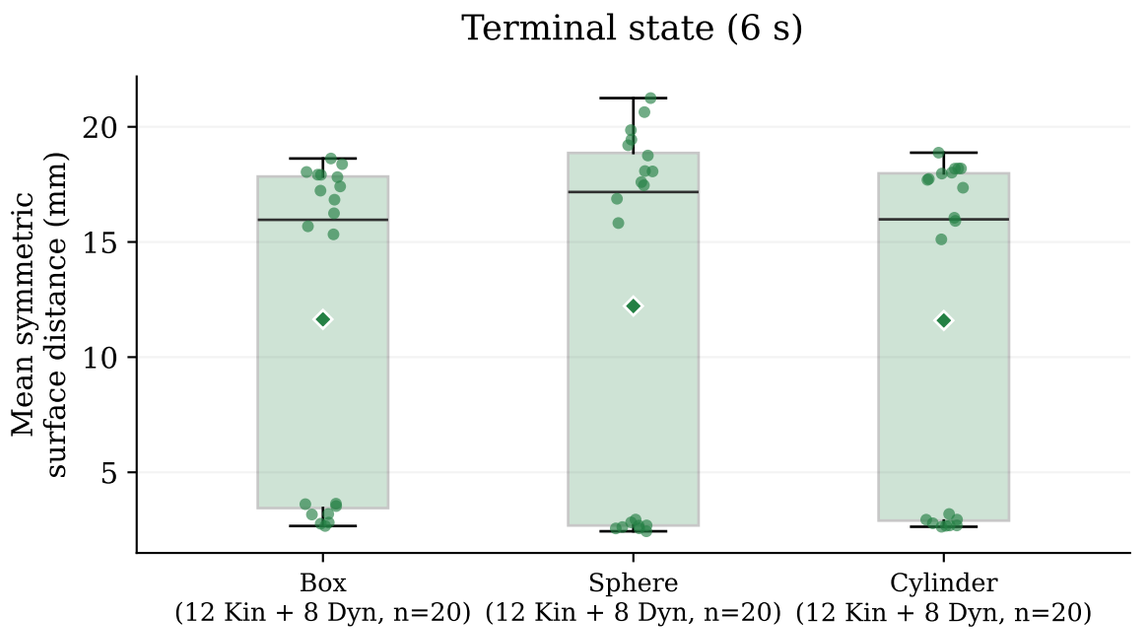
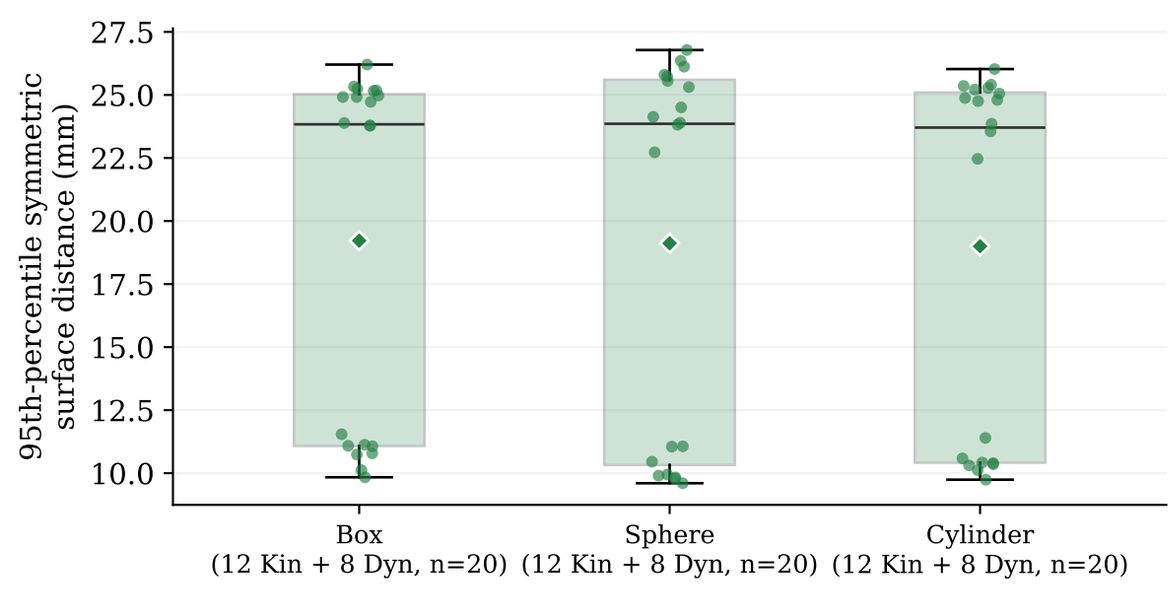
Group A: By Interacting Object (Box, Sphere, Cylinder) — Motions: Kinematic (Horizontal, Vertical, Circle, Hold) & Dynamic (Drop, Slide, Throw, Tumble)







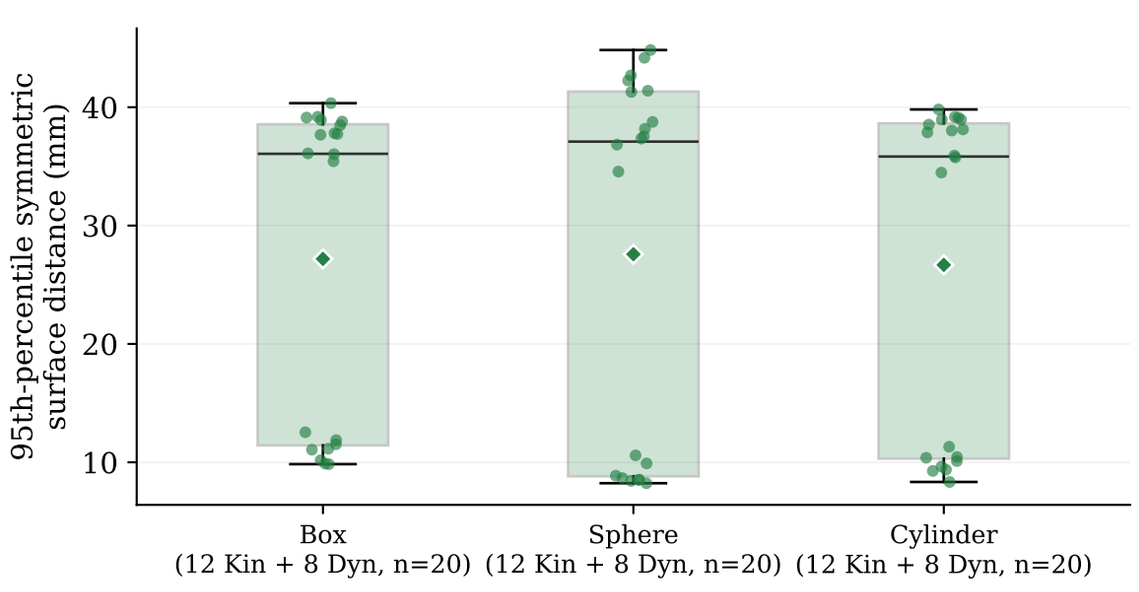


Figure 2a: Frozen object-model checkpoint 10,800. Elementary objects (Box, Sphere, Cylinder) across 12 Kinematic (Horizontal, Vertical, Circular, Hold) and 8 Dynamic (Drop, Slide, Throw, Tumble) trajectories per shape. Rows show mean and 95th-percentile symmetric surface distance on a 0.02 m measurement grid.